

Living Machines

Adopting nature's designs to advance human technology

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Technology is not necessarily only physics, chemistry or biology but a combination of the three. However it is from nature that most man-made inventions draw inspiration. Why re-invent the wheel when nature has done it for us? We don't even need to understand a phenomenon completely to use it to our advantage. The design exists, and deciphering is optional. In some cases, the deciphering of nature's mechanisms happens first and is only then applied to the appropriate areas of need. Other times, we can directly apply the affects without having gathered all the knowledge. Either way, we may like to imagine we are inventing the fruit, when we are merely picking it from the tree. Though God or (insert higher power here) granted us the metaphorical fruit to find, humans can certainly take pride in inventing the salads and the smoothies!

Evaporation engines

The water cycle is a long standing natural process almost as old as the earth itself. We have developed a few ways to tap into the movement of this resource as it falls downward (hydro-power at river dams) or flows sideways (tidal energy), but the significant other half of the cycle is when the water goes



The making of a moisture mill

back up into the atmosphere as water vapour. This ubiquitous process takes place over all water bodies across the world and as such, is a vast untapped resource. To draw power from this process, scientists and researchers at Columbia University have utilised the natural physical response of *Bacillus* spores to change in humidity. *Bacillus* spores are inert cells, a form of dormant life, yet they exhibit a physical response to chemical stimuli. Based on changes in relative humidity, the spore cells expand and contract, undergoing a change in diameter of as much as 12%. The researchers deposit these spores on a thin tape-like rubber film, turning it into a stimuli responsive material which in turn rolls up or straightens

out depending on the surrounding water content. The spores are self assembling so there is no special procedure for alignment of the spores in the film during fabrication. This is how the natural biological mechanism is converted into a humidity sensing actuator which can do work. The motion can also be converted to electricity by using it to move a coil in a magnetic field. Although the precise proteins and exact mechanisms involved in this feat are still being investigated, it goes to show that an understanding of the underlying principles of the universe aren't necessary to harness its energy, or to be published in *Nature*. This new class of energy harnessing is called hygrovoltaics. The group responsible